

AQUABC v0.2 — Ecological Model Reference Manual

ESTAS-AQUABC Development Team

February 2026

Contents

1	Introduction	3
1.1	Model Architecture	3
1.2	Source Code Organisation	3
2	Pelagic Model	4
2.1	State Variables	4
2.2	Phytoplankton Submodels	4
2.2.1	Temperature Limitation (CTMI)	4
2.2.2	Light Limitation	5
2.2.3	Nutrient Limitation (Synthesizing Unit)	5
2.2.4	Growth, Respiration, Mortality	5
2.3	Zooplankton Dynamics	5
2.4	Organic Matter Cycling	5
2.4.1	POM Dissolution	5
2.4.2	DOM Mineralisation	6
2.5	Dissolved Oxygen	6
2.6	Nutrient Cycling	6
2.7	Redox Chemistry	6
3	Sediment Diagenesis Model	7
3.1	Overview	7
3.2	Sediment State Variables (per layer)	7
3.3	Transport Processes	7
3.3.1	Advection	7
3.3.2	Diffusion	7
3.3.3	Particle Mixing (Bioturbation)	7
3.3.4	Burial	8
3.3.5	Combined Transport	8
3.4	Erosion and Deposition	8
3.5	Sediment–Water Fluxes	8
3.6	Bioturbation and Bioirrigation	8
3.6.1	Biodiffusion Coefficient	8
3.6.2	Bioturbation Parameters	9
3.6.3	Bioirrigation	9
3.6.4	Lower Boundary Condition	9

3.7	Sediment pH Correction	9
4	CO₂ System	11
5	Macroalgae Model	12
5.1	Droop Quota Nutrient Limitation	12
5.2	Growth	12
6	Allelopathy Module	13
7	Model Constants	14
7.1	Pelagic Model Constants (318 parameters)	14
7.2	Sediment Model Constants (170 parameters)	14
8	Numerical Methods	15
8.1	Safety Functions	15
8.2	Time Integration	15
8.3	Mass Balance Safeguards	15
9	References	16

1 Introduction

AQUABC v0.2 is a comprehensive aquatic biogeochemical model that simulates coupled pelagic–sediment dynamics in lakes, reservoirs, estuaries, and coastal waters. The model is designed to be embedded in hydrodynamic transport frameworks (such as ESTAS or SHYFEM) that provide advection, dispersion, and external forcing.

1.1 Model Architecture

AQUABC consists of five coupled sub-models:

1. **Pelagic model:** 32 state variables covering phytoplankton (4 groups), zooplankton, organic matter, nutrients, dissolved oxygen, carbonate chemistry, and redox-sensitive metals.
2. **Sediment diagenesis model:** 24 state variables per sediment layer for early diagenesis with full redox sequencing and bioturbation.
3. **Macroalgae model:** 6 state variables using Droop quota kinetics.
4. **Allelopathy module:** 4 secondary metabolite pools for inter-species chemical inhibition.
5. **CO₂SYS:** Full carbonate chemistry solver (CDIAC implementation).

1.2 Source Code Organisation

Component	File
Pelagic kinetics	aquabc_II_pelagic_model.f90
Phytoplankton libraries	aquabc_II_pelagic_lib_*.f90
Sediment model	aquabc_II_sediment_model_1_fast.f90
Bioturbation module	aquabc_II_sediment_bioturbation.f90
Sediment model constants	aquabc_II_sediment_model_constants.f90
CO ₂ SYS	aquabc_II_co2sys.f90
Macroalgae	mod_MACROALGAE.f90
Allelopathy	mod_ALLELOPATHY.f90
Precision definitions	precision_kinds.f90
Parameter dictionary	STRING_UTILS.f90 (para_aqua module)

2 Pelagic Model

2.1 State Variables

The pelagic model tracks 32 state variables in each computational box:

Index	Variable	Unit	Description
1	NH ₄ -N	mg/L	Ammonium nitrogen
2	NO ₃ -N	mg/L	Nitrate nitrogen
3	PO ₄ -P	mg/L	Orthophosphate phosphorus
4	DOXY	mg/L	Dissolved oxygen
5	DIA-C	mg/L	Diatom carbon
6	CYN-C	mg/L	Non-fixing cyanobacteria carbon
7	OPA-C	mg/L	Other planktonic algae carbon
8	FIX-CYN-C	mg/L	N-fixing cyanobacteria carbon
9	ZOO-C	mg/L	Zooplankton carbon
10	DET-C	mg/L	Detrital organic carbon
11	DET-N	mg/L	Detrital organic nitrogen
12	DET-P	mg/L	Detrital organic phosphorus
13	DOC	mg/L	Dissolved organic carbon
14	DON	mg/L	Dissolved organic nitrogen
15	DOP	mg/L	Dissolved organic phosphorus
16	DSi	mg/L	Dissolved silica
17	PSi	mg/L	Particulate (biogenic) silica
18	DIC	mg/L	Dissolved inorganic carbon
19	ALK	meq/L	Alkalinity
20	Mn(II)	mg/L	Dissolved manganese
21	Mn(IV)	mg/L	Particulate manganese oxide
22	Fe(II)	mg/L	Dissolved ferrous iron
23	Fe(III)	mg/L	Particulate ferric iron
24	S(-II)	mg/L	Dissolved sulphide
25	S(VI)	mg/L	Dissolved sulphate
26	CH ₄	mg/L	Dissolved methane
27–28	NOST-VEG, NOST-HET	mg/L	Nostocales vegetative cells and heterocysts
29–32	SEC-METAB 1–4	mg/L	Allelopathic secondary metabolites

2.2 Phytoplankton Submodels

Four phytoplankton groups share a common kinetic framework:

2.2.1 Temperature Limitation (CTMI)

The Cardinal Temperature Model with Inflection (Rosso et al., 1993):

$$\mu(T) = \begin{cases} \mu_{\max} \cdot \frac{(T - T_{\max})(T - T_{\min})^2}{(T_{\text{opt}} - T_{\min}) [(T_{\text{opt}} - T_{\min})(T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max})(T_{\text{opt}} + T_{\min} - 2T)]} & T_{\min} < T < T_{\max} \\ 0 & \text{otherwise} \end{cases}$$

2.2.2 Light Limitation

Steele/Smith formulation with optional Platt-style photoinhibition:

$$f_I = \frac{I}{I_s} \exp\left(1 - \frac{I}{I_s}\right) \cdot \exp(-\beta \cdot I)$$

where I is the depth-averaged PAR (Photosynthetically Active Radiation), I_s is the light saturation parameter, and β controls photoinhibition intensity.

2.2.3 Nutrient Limitation (Synthesizing Unit)

The Synthesizing Unit colimitation model (Saito et al., 2008) replaces Liebig's minimum:

$$f_N = \frac{1}{\frac{1}{f_{N\text{-lim}}} + \frac{1}{f_{P\text{-lim}}} + \frac{1}{f_{Si\text{-lim}}} - \frac{1}{f_{N\text{-lim}} \cdot f_{P\text{-lim}}} - \frac{1}{f_{N\text{-lim}} \cdot f_{Si\text{-lim}}} - \frac{1}{f_{P\text{-lim}} \cdot f_{Si\text{-lim}}} + \frac{1}{f_{N\text{-lim}} \cdot f_{P\text{-lim}} \cdot f_{Si\text{-lim}}}}$$

where each individual limitation follows Monod kinetics: $f_{N\text{-lim}} = [\text{DIN}] / ([\text{DIN}] + K_{\text{HS},N})$.

2.2.4 Growth, Respiration, Mortality

$$\frac{d[\text{PHYTO}]}{dt} = (\mu - R - D) \cdot [\text{PHYTO}]$$

where μ is the temperature-, light-, and nutrient-limited growth rate, R is Arrhenius-corrected respiration $R = k_{R,20} \cdot \theta^{T-20}$, and D is mortality (enhanced under hypoxia).

2.3 Zooplankton Dynamics

Zooplankton feeding uses the Active Switching Model (Gentleman et al., 2003):

$$I_i = I_{\max} \cdot \frac{\rho_i \cdot [F_i] / \sum_j (\rho_j \cdot [F_j])}{K_F + \sum_j [F_j]} \cdot \sum_j [F_j]$$

where I_i is ingestion rate of food source i , ρ_i are preference weights, $[F_i]$ are food concentrations, and K_F is the half-saturation constant.

2.4 Organic Matter Cycling

2.4.1 POM Dissolution

Particulate organic matter (C, N, P, Si) dissolves into dissolved forms:

$$R_{\text{diss}} = k_{\text{diss},20} \cdot \theta^{T-20} \cdot \frac{[\text{POM}]}{[\text{POM}] + K_{\text{HS}}} \cdot [\text{POM}]$$

2.4.2 DOM Mineralisation

DOC mineralisation follows a six-pathway redox sequence, each with its own rate constant, temperature correction, and half-saturation kinetics:

1. Aerobic respiration (O_2)
2. Denitrification (NO_3^-)
3. Manganese reduction ($Mn(IV)$)
4. Iron reduction ($Fe(III)$)
5. Sulphate reduction ($S(VI)$)
6. Methanogenesis (DOC as terminal acceptor)

Each pathway is regulated by:

- Substrate half-saturation: $f = [DOC]/([DOC] + K_{HS,DOC})$
- Electron acceptor availability: $f = [EA]/([EA] + K_{HS,EA})$
- Inhibition by higher-priority acceptors (reversed Monod)
- pH correction (Gaussian)

2.5 Dissolved Oxygen

$$\frac{d[O_2]}{dt} = R_{\text{reaeration}} + R_{\text{photosynthesis}} - R_{\text{respiration}} - R_{\text{nitrification}} - R_{\text{oxidation}}$$

Reaeration follows the O'Connor–Dobbins or Wanninkhof formulation depending on wind speed input.

2.6 Nutrient Cycling

- **Nitrification:** $NH_4^+ \rightarrow NO_3^-$ (Monod kinetics in O_2 and NH_4 , pH-corrected)
- **Denitrification:** $NO_3^- \rightarrow N_2$ (only under low- O_2 conditions)
- **Phosphate speciation:** pH-dependent sorption with $Fe(III)$

2.7 Redox Chemistry

Full thermodynamic hierarchy: $O_2 > NO_3^- > Mn(IV) > Fe(III) > SO_4^{2-} > \text{methanogenesis}$.

Metal speciation (Fe, Mn) includes oxidation–reduction kinetics and equilibrium solubility calculations.

3 Sediment Diagenesis Model

3.1 Overview

The sediment model tracks 24 state variables per layer in a multi-layer vertical discretisation. It shares the same biogeochemical reaction framework as the pelagic model (POM dissolution, DOM mineralisation, nitrification, denitrification, full redox sequence) applied to sediment porewater and solid phases.

File: `aquabc_II_sediment_model_1_fast.f90`

3.2 Sediment State Variables (per layer)

The 24 sediment state variables mirror the pelagic model with sediment-specific organic matter pools, dissolved nutrients, and redox species. Each variable exists in dissolved (porewater) and/or particulate (solid) phase.

3.3 Transport Processes

Four transport mechanisms operate on each sediment layer:

3.3.1 Advection

Porewater flow between layers:

$$R_{\text{adv,in}} = \frac{C_{\text{entering}}}{\max(\phi, 10^{-20})} \cdot |v_{\text{adv}}|$$

$$R_{\text{adv,out}} = \frac{C_{\text{current}} \cdot f_{\text{sol}}}{\max(\phi, 10^{-20})} \cdot |v_{\text{adv}}|$$

where ϕ is porosity and f_{sol} is the dissolved fraction.

3.3.2 Diffusion

Concentration gradient-driven with tortuosity correction:

$$R_{\text{diff}} = \frac{D_{\text{eff}}}{1 + 3(1 - \phi)} \cdot \frac{\Delta C}{\max(\ell, 10^{-20})}$$

where $\ell = 0.5 \cdot (h_{i-1} + h_i)$ is the mixing length between adjacent layers.

3.3.3 Particle Mixing (Bioturbation)

For solid-phase species:

$$R_{\text{pmix}} = \frac{D_b \cdot \Delta C_{\text{solid}}}{\max(\ell, 10^{-20})}$$

where D_b is the biodiffusion coefficient (see §3.5).

3.3.4 Burial

Permanent removal to deep sediment:

$$R_{\text{burial}} = \frac{C \cdot w_{\text{burial}}}{\max(h, 10^{-20})}$$

where w_{burial} is the burial rate (m/day) and h is the layer thickness.

3.3.5 Combined Transport

$$\left. \frac{dC}{dt} \right|_{\text{transport}} = R_{\text{diff}} \cdot \phi + R_{\text{burial}} + R_{\text{pmix}} + R_{\text{adv,in}} - R_{\text{adv,out}}$$

3.4 Erosion and Deposition

When the sediment transport model is active (`isedi` > 0), layer concentrations are adjusted for sediment–water interface movement:

- **Deposition** ($H_{\text{erodep}} \leq 0$): Upper layer material mixed into lower layers
- **Erosion** ($H_{\text{erodep}} > 0$): Lower layer material exposed at the surface

In ESTAS standalone mode, `isedi` = 0 (prescribed settling used instead).

3.5 Sediment–Water Fluxes

24 dissolved fluxes are exchanged at the sediment–water interface based on concentration gradients between porewater and overlying water. Positive flux = release from sediment to water column.

Flux Index	Species	Direction
1	NH ₄ -N	Typically upward (nutrient release)
2	NO ₃ -N	Variable (nitrification/denitrification)
3	PO ₄ -P	Typically upward
4	DOXY	Typically downward (sediment O ₂ demand)
9	DOC	Typically upward
15	ALK	Typically upward

3.6 Bioturbation and Bioirrigation

File: `aquabc_II_sediment_bioturbation.f90`

When enabled (`switch_partmixing` = 1, default), particle mixing coefficients are dynamically computed based on local conditions rather than using a uniform constant.

3.6.1 Biodiffusion Coefficient

$$D_b(z, t) = D_{b0} \cdot \exp\left(-\frac{z}{z_{\text{mix}}}\right) \cdot \frac{[\text{O}_2]}{[\text{O}_2] + K_{\text{HS},\text{O}_2}} \cdot \left[1 + A \cos\left(\frac{2\pi(d - d_{\text{peak}})}{365}\right)\right]$$

The three multiplicative factors represent:

1. **Depth attenuation:** Exponential decay reflecting the decrease of macrofauna density with depth in sediments (Boudreau, 1997).
2. **Oxygen dependence:** Monod kinetics reflecting that benthic fauna require oxygen; under anoxia, bioturbation ceases.
3. **Seasonal modulation:** Sinusoidal annual cycle with summer peak, reflecting temperature-driven activity patterns (Soetaert et al., 1996).

3.6.2 Bioturbation Parameters

Parameter	Symbol	Default	Unit
Surface biodiffusion	D_{b0}	from input	m^2/day
Mixing depth	z_{mix}	0.05	m
O ₂ half-saturation	K_{HS,O_2}	2.0	mg/L
Seasonal amplitude	A	0.3	—
Peak day	d_{peak}	200	day
Irrigation enhancement	α_0	3.0	—
Irrigation depth	z_{irr}	0.04	m

3.6.3 Bioirrigation

Burrow-dwelling organisms pump overlying water through the sediment, enhancing porewater solute exchange beyond molecular diffusion:

$$\alpha(z, t) = 1 + \alpha_0 \cdot \exp\left(-\frac{z}{z_{\text{irr}}}\right) \cdot \frac{[\text{O}_2]}{[\text{O}_2] + K_{\text{HS},\text{O}_2}} \cdot f_{\text{season}}(t)$$

The irrigation factor α multiplies the molecular diffusion coefficient for dissolved-phase species:

$$D_{\text{eff,enhanced}} = \alpha \cdot D_{\text{eff,molecular}}$$

Bioirrigation is only applied to dissolved and mixed-phase species (not purely particulate). It is controlled by `switch_bioirrigation` (default = 1, enabled).

3.6.4 Lower Boundary Condition

A zero-flux (Neumann) boundary condition at the bottom of the deepest layer:

$$\left. \frac{\partial C}{\partial z} \right|_{z_{\text{bottom}}} = 0 \quad \Rightarrow \quad R_{\text{pmix}}(N) = -R_{\text{pmix,in}}(N)$$

This prevents artificial mass loss through the lower boundary.

3.7 Sediment pH Correction

Gaussian formulation (different from pelagic exponential):

$$f_{\text{pH}} = \exp\left(-\frac{(\text{pH} - \text{pH}_{\text{opt}})^2}{2\sigma^2}\right)$$

where $\text{pH}_{\text{opt}} = (\text{pH}_{\text{min}} + \text{pH}_{\text{max}})/2$ and $\sigma = (\text{pH}_{\text{max}} - \text{pH}_{\text{min}})/4$.

4 CO₂ System

The full carbonate chemistry system follows the CDIAC CO2SYS implementation, computing equilibrium speciation of dissolved inorganic carbon:



Given any two of (pH, pCO₂, DIC, ALK, CO₃²⁻), the system solves for all others using iterative Newton–Raphson methods with temperature- and salinity-dependent equilibrium constants.

5 Macroalgae Model

5.1 Droop Quota Nutrient Limitation

$$Q_N = \frac{[\text{MAC-N}]}{[\text{MAC-C}]}, \quad \Phi_N = \max\left(0, 1 - \frac{Q_{0,N}}{Q_N}\right)$$

5.2 Growth

$$R_{\text{growth}} = k_{G,20} \cdot \theta^{T-20} \cdot \Phi_N \cdot \Phi_P \cdot \Phi_L \cdot \Phi_S \cdot [\text{MAC-C}]$$

where $\Phi_S = 1 - [\text{MAC-C}]/C_{\text{max}}$ is the logistic space limitation.

6 Allelopathy Module

Four secondary metabolite pools model inter-species chemical inhibition. Each phytoplankton group produces a specific metabolite that inhibits growth of competing groups via Monod-type dose-response:

$$f_{\text{inhib}} = 1 - \frac{[\text{SEC-METAB}]}{[\text{SEC-METAB}] + K_{\text{HS,inhib}}}$$

7 Model Constants

7.1 Pelagic Model Constants (318 parameters)

Organised in 15 categories:

1. **General** (1–5): Temperature ranges, O₂ tolerances
2. **Diatoms** (5–31): Growth, respiration, mortality, nutrient kinetics, Si:C ratio
3. **Non-fixing cyanobacteria** (29–50): Buoyancy regulation, light adaptation
4. **N-fixing cyanobacteria** (51–74): Nitrogen fixation rates, heterocyst parameters
5. **Other planktonic algae** (75–96): Growth and loss parameters
6. **Zooplankton** (97–133): Feeding preferences, active switching parameters
7. **POM dissolution** (134–145): Rate constants per organic matter type
8. **DOM mineralisation** (146–152): Aerobic pathway parameters
9. **Nitrification** (152–157): Rate and half-saturation constants
10. **Redox chemistry** (158–209): All six electron acceptor pathways
11. **Methane** (210–234): CH₄ oxidation and production
12. **Settling** (235–250): Settling velocities per state variable
13. **pH correction** (251–266): Optimal pH ranges per process
14. **Dissolved metals** (267–276): Fe/Mn oxidation and dissolution rates
15. **Nostocales** (276–298): Heterocyst-forming cyanobacteria parameters

All constants are read from an external file and stored in the `para_aqua` name–value dictionary for runtime access.

7.2 Sediment Model Constants (170 parameters)

Sediment-specific rate constants mirror the pelagic model with separate values optimised for sediment conditions (lower temperatures, higher organic matter concentrations, different pH ranges). Read from `BOTTOM_SEDIMENT_MODEL_INPUT.txt` via the `WCONST` constants file.

8 Numerical Methods

8.1 Safety Functions

All division operations use `max(divisor, 10^{-20})` floors to prevent NaN/Inf. Temperature-dependent exponentials use `safe_exp()` with overflow protection. pH is clamped to [4, 11].

8.2 Time Integration

The host transport framework (ESTAS or SHYFEM) provides the numerical time integration. AQUABC returns instantaneous rate derivatives which are integrated by the host using Euler or second-order Runge–Kutta methods.

8.3 Mass Balance Safeguards

- Maximum 50% loss per timestep for zooplankton and phytoplankton
- Concentration floors at 0 (non-negative constraint)
- Derivative clamping to prevent numerical overshoot

9 References

- Boudreau, B.P. (1997). *Diagenetic Models and Their Implementation*. Springer.
- Gentleman, W., Leising, A., Frost, B., Strom, S., and Murray, J. (2003). Functional responses for zooplankton feeding on multiple resources. *J. Plankton Res.* 25:1215–1234.
- Middelburg, J.J., Soetaert, K., and Herman, P.M.J. (1997). Empirical relationships for use in global diagenetic models. *Deep-Sea Res. I* 44:327–344.
- Morgan, B. and Lahav, O. (2007). The effect of pH on the kinetics of spontaneous Fe(II) oxidation. *Chemosphere*.
- Platt, T., Gallegos, C.L., and Harrison, W.G. (1980). Photoinhibition of photosynthesis in natural assemblages of marine phytoplankton. *J. Mar. Res.* 38:687–701.
- Rosso, L., Lobry, J.R., and Flandrois, J.P. (1993). An unexpected correlation between cardinal temperatures of microbial growth highlighted by a new model. *J. Theor. Biol.* 162:447–463.
- Saito, M.A., Goepfert, T.J., and Ritt, J.T. (2008). Some thoughts on the concept of colimitation. *Limnol. Oceanogr.* 53:276–290.
- Soetaert, K., Herman, P.M.J., and Middelburg, J.J. (1996). A model of early diagenetic processes from the shelf to abyssal depths. *Geochim. Cosmochim. Acta* 60:1019–1040.
- Steele, J.H. (1962). Environmental control of photosynthesis in the sea. *Limnol. Oceanogr.*
- Stumm, W. and Morgan, J.J. (1996). *Aquatic Chemistry*. Wiley.
- CDIAC CO2SYS program documentation.
- EPA-829-R-14-007. AQUATOX Technical Documentation.